

Install Docker in device through Terminal

```
priyansh@IDEAPAD-18UT:~$ sudo apt update
sudo apt install -y ca-certificates curl gnupg
[sudo] password for priyansh:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components
[21.9 kB]
```

Add Docker GPG Key and Repo

```
priyansh@IDEAPAD-18UT:~$ sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
  sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
priyansh@IDEAPAD-18UT:~$ echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker
.gpg] \
  https://download.docker.com/linux/ubuntu \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
priyansh@IDEAPAD-18UT:~$ sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-compose-plu
gin
```

Run without sudo

```
priyansh@IDEAPAD-18UT:~$ sudo usermod -aG docker $USER
newgrp docker
```

Verify Installation

```
priyansh@IDEAPAD-18UT:~$ docker info
docker --version
Client: Docker Engine - Community
Version: 29.4.1
Context: default
Debug Mode: false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.33.0
    Path: /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version: v5.1.3
    Path: /usr/libexec/docker/cli-plugins/docker-compose

Server:
Containers: 0
  Running: 0
  Paused: 0
  Stopped: 0
Images: 0
Server Version: 29.4.1
Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
  Volume: local
```

Pull and Run Official Images

```
Ubuntu
Docker version 29.4.1, build 055a478
priyansh@IDEAPAD-18UT:~$ # Pull nginx and Ubuntu images
docker pull nginx
docker pull ubuntu

# List downloaded images
docker images
Using default tag: latest
latest: Pulling from library/nginx
801a1ad15b4e: Pull complete
85c66128325a: Pull complete
4677c2a9a3d4: Pull complete
3531af2bc2a9: Pull complete
ff048f1f2159: Pull complete
ce776bbcd0d: Pull complete
677c63196868: Pull complete
54c38c75806e: Download complete
69989ccd189b: Download complete
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
Using default tag: latest
latest: Pulling from library/ubuntu
b40150c1c271: Pull complete
92842f25412d: Download complete
Digest: sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41c7b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

In Use
IMAGE          ID          DISK USAGE  CONTENT SIZE  EXTRA
nginx:latest    6e23479198b9  240MB       65.8MB
ubuntu:latest   c4a8d5503dfb  119MB       31.7MB
```

Run nginx container and explore it

```
priyansh@IDEAPAD-18UT:~$ docker run -d -p 8080:80 --name my-nginx nginx
4d0a895daee4744c3c648551af14dc3b4c268ec8f93ad009f35d956f88e04f77
priyansh@IDEAPAD-18UT:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS      PORTS
4d0a895daee4   nginx     "/docker-entrypoint..." 42 seconds ago   Up         0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
priyansh@IDEAPAD-18UT:~$ docker exec -it my-nginx bash
root@4d0a895daee4:/# ls /usr/share/nginx/html/
50x.html index.html
root@4d0a895daee4:/# exit
exit
```

Stop Docker and remove it

```
priyansh@IDEAPAD-18UT:~$ docker stop my-nginx
my-nginx
priyansh@IDEAPAD-18UT:~$ docker rm my-nginx
my-nginx
priyansh@IDEAPAD-18UT:~$
```